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## **O-RAN Work Group 10 (OAM for O-RAN) Cloud Native OAM Report**

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## Foreword

This Technical Report (TR) has been produced by WG10 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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## Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

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## Executive summary

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## Introduction

A key purpose of standardization is to enable multivendor interoperability, i.e., to allow implementations from different vendors to work together. In particular, standardization of management interfaces should simplify management of mobile networks (e.g. 3GPP TS 32.101 [i.12]). In 3GPP SA5, the concept of management services is used as described in TS 28.533 [i.4].

To support interoperability, it is fundamental that (Management) Service Producer and (Management) Service Consumer(s) agree on use cases and requirements. Then, based on use cases and requirements, a model of the information to be managed must be agreed between producer and consumers and according to those, a specific solution set, in terms of protocol and data model, must be also defined and agreed.

3GPP uses a phased approach defined in 21.900. In 3GPP SA5, in general, Stage 1 specifications define requirements and use cases. Stage 2 specification define the managed entities in terms of a common information model that consumers and producers shall agree on and exchange to realize the use cases defined in stage 1. Stage 3 specification defines specific solution sets that implement the models defined at stage 2.

The service-based management architecture in TS 28.533 [i.4] defines a management service to be composed of management operations and/or notifications (referred to as Management Service Component Type A), data to be

managed (referred to as Management Service Component Type B) and fault and performance information (referred to as Management Service Component Type C). The latter component is optional in the definition of a management service. The O-RAN O1 specification [i.1] largely consists of references to Stage 2 and Stage 3 instantiations of these three component types.

In this report, we visit each of these component types and examine whether emerging cloud-native management design patterns might motivate the creation of new Stage 1 requirements and use cases, new Stage 2 operations and information models, and new Stage 3 protocols and transport mechanisms. We begin in clause 4 with Management Component Type B, as information and data models are foundational to our study. Then clause 5 examines Management Component Type A, the set of operations and notifications defined to work on the data. clause 6 examines the control aspects of Management Component Type C and clause 7 examines transport of those fault, performance, and trace information from a cloud-native perspective.

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## 1 Scope

The objective of this Technical Report is to study the data formats, management operations, and telemetry transport mechanisms for OAM from a cloud-native perspective. Where applicable, the present document proposes recommendations or potential solutions for cloud-native OAM.

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## 2 References

### 2.1 Normative references

Not applicable.

### 2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

- [i.1] O-RAN.WG10.O1-Interface: "O-RAN Operations and Maintenance Interface Specification".
- [i.2] ITU-T Recommendation X.500 (1993): "Information technology - Open Systems Interconnection - The Directory: Overview of concepts, models and services".
- [i.3] 3GPP TS 28.532 v17.3.0: "Management and orchestration; Generic management services".
- [i.4] 3GPP TS 28.533 v17.3.0: "Management and orchestration; Architecture framework".
- [i.5] 3GPP TS 28.541 v17.9.0: "Management and orchestration of 5G networks; Network Resource Model (NRM); Stage 2 and stage 3".
- [i.6] 3GPP TS 28.622 v17.5.0: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".
- [i.7] 3GPP TS 28.623 v17.5.1: "Generic Network Resource Model (NRM); Integration Reference Point (IRP); Solution Set (SS) definitions".
- [i.8] 3GPP TS 32.156 v17.2.0: "Telecommunication management; Fixed Mobile Convergence (FMC); Model repertoire".

- [i.9] 3GPP TS 32.160 v17.6.0: “Management and orchestration; Management service template”.
- [i.10] 3GPP TS 32.300 v17.0.0: “Telecommunication management; Configuration Management (CM); Name convention for managed objects”.
- [i.11] 3GPP TS 21.900 v17.2.0: “Technical Specification Group Services and System Aspects; Technical Specification Group working methods”.
- [i.12] 3GPP TS 32.101 v17.0.0: “Telecommunication management; Principles and high level requirements”.
- [i.13] O-RAN.WG10.OAM-Architecture: “O-RAN Operations and Maintenance Architecture”.
- [i.14] O-RAN.WG5.O-DU-O1: “O1 Interface specification for O-DU”.
- [i.15] 3GPP TS 28.550 v18.2.0: “Management and orchestration; Performance assurance”.
- [i.16] 3GPP TS 28.621 v17.0.0: “Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Requirements”.
- [i.17] 3GPP TS 32.401 v17.0.0: “Telecommunication management; Performance Management (PM); Concept and requirements”.
- [i.18] 3GPP TS 32.404 v17.0.0: “Telecommunication management; Performance Management (PM); Performance measurements; Definitions and template”.
- [i.19] 3GPP TS 32.421 v17.4.0: “Telecommunication management; Subscriber and equipment trace; Trace concepts and requirements”.
- [i.20] 3GPP TS 28.552 v17.11.0: “Management and orchestration; 5G performance measurements”.
- [i.21] 3GPP TS 28.111 v18.1.0: “Management and orchestration; Fault Management (FM)”.
- [i.22] O-RAN.WG10.Information Model and Data Models: “O-RAN Information Model and Data Models Specification”.
- [i.23] O-RAN.WG1.Use-Cases-Analysis-Report: “O-RAN Working Group 1 Use Cases Analysis Report”.
- [i.24] O-RAN.WG1.Slicing-Architecture: “O-RAN Work Group 1 (Use Cases and Overall Architecture) Slicing Architecture”.
- [i.25] 3GPP TR 21.905 V17.2.0: "Vocabulary for 3GPP Specifications".
- [i.26] 3GPP TS 32.111-1 v17.0.0: "Part 1: 3G fault management requirements".
- [i.27] 3GPP TS 32.423 v18.0.0: "Subscriber and equipment trace; Trace data definition and management".
- [i.28] O-RAN TS O-RAN.WG10.PMeas: "O-RAN O1 Performance Measurements Specification".
- [i.29] 3GPP TS 28.537 v18.2.0: “Management and orchestration; Management capabilities”.
- [i.30] 3GPP TS 32.432 v18.0.0: “Telecommunication management; Performance measurement: File format definition”.
- [i.31] O-RAN-WG1.Decoupled-SMO-Architecture: “O-RAN Work Group 1 (Use Cases and Overall Architecture) Decoupled SMO Architecture”

## 3 Definition of terms, symbols and abbreviations

### 3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [i.25] and the following terms apply:

**Telemetry:** The process of gathering data of a system during execution and communicating it to a remote location.

NOTE: Telemetry data can be used to understand the behavior and performance of the system.

## 3.2 Symbols

Void

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [i.1] and the following apply:

|         |                                                                   |
|---------|-------------------------------------------------------------------|
| 3GPP    | 3rd Generation Partnership Project                                |
| ASN.1   | Abstract Syntax Notation One                                      |
| CI/CD   | Continuous Integration, Continuous Delivery/Continuous Deployment |
| CM      | Configuration Management                                          |
| DN      | Distinguished Name                                                |
| FM      | Fault Management                                                  |
| HTTP    | HyperText Transfer Protocol                                       |
| IOC     | Information Object Class                                          |
| IRP     | Integration Reference Point                                       |
| JSON    | JavaScript Object Notation                                        |
| MnS     | Management Service                                                |
| MOC     | Managed Object Class                                              |
| MOI     | Managed Object Instance                                           |
| NETCONF | NETwork CONfiguration protocol                                    |
| NRM     | Network Resource Model                                            |
| PM      | Performance Management or Performance Measurements                |
| RFC     | Request For Comments                                              |
| SBMA    | Service Based Management Architecture                             |
| SMO     | Service Management and Orchestration                              |
| SMOS    | SMO Service                                                       |
| SSH     | Secure Shell                                                      |
| TLS     | Transport Layer Security                                          |
| UML     | Unified Modeling Language                                         |
| XML     | eXtensible Markup Language                                        |
| YAML    | Yet Another Markup Language                                       |
| YANG    | Yet Another Next Generation                                       |



## 4 Analysis of Data Format

### 4.1 Background

#### 4.1.1 MnS Component Type B

A Management Service Component Type B is defined in clause 4.2.3.1 of 3GPP TS 28.533 [i.4] as referring to management information represented by information models representing the managed entities. The definition goes on to state that the terminology “MnS component type B” is interchangeable with the term “Network Resource Model (NRM)”. Two examples of NRMs are given, one being the generic NRM 3GPP TS 28.622 [i.6] and one being the 5G NRM 3GPP TS 28.541 [i.5].

3GPP TS 32.156 [i.8] defines a model repertoire for building abstract information models of network resources. It outlines a subset of the Unified Modeling Language (UML) to be used in model development. It defines the stereotype Information Object Class (IOC) as a key building block for information models. To define a new IOC, a modeler gives it a class name and an associated table of attributes, each of which can take on a range of usually primitive or simple compound values according to its data type. The repertoire also defines the <<names>> stereotype association which allows one class to name-contain other classes. The name-containment relationship can be single (represented with [ParentIOC] 1 <<names>> 0..1 [Child IOC] in the UML diagram) or multiple (represented with [ParentIOC] 1 <<names>> \* [ChildIOC] in the UML diagram). Typically, when a 3GPP document defines a collection of IOCs, the name-containment relationships are specified with a collection of diagrams in a separate clause of the document from the per-IOC clauses that define the attributes of each new class.

References to specific object instances are implemented with the concept of a Distinguished Name (DN) from the X.500 family of specifications [i.2]. References to specific object instances are important to express a non-containment association from one object instance to another, as well as to indicate which object is the subject of a management operation as described in clause 5. A DN as used by the name convention for managed objects 3GPP TS 32.300 [i.10] is a sequence of Attribute Type, Attribute Value pairs; the Attribute Type is a string containing the class name of the ChildIOC, and the AttributeValue is a string containing the value of the designated “naming attribute” of the ChildIOC. This “naming attribute” is usually called “id”.

#### 4.1.2 Current Solution Set Methodology for Component Type B

To enable actual management of a network resource, the abstract information model must be converted into a Stage 3 specification that enables interoperable implementations of management operations on the modeled data. Several so-called Solution Sets have been defined by 3GPP, including based on CORBA, NETCONF/YANG, and OpenAPI [i.aaa, i.ee]. Below we outline the conventions in the 3GPP management services template [i.eee] for OpenAPI/JSON schema and NETCONF/YANG even if current O1 specification [i.1] mandates the use of the YANG Stage 3 solution set rather than the OpenAPI/YAML Stage 3 solution set.

##### 4.1.2.1 OpenAPI/JSON Schema Data Models

A fragment of an example OpenAPI/JSON Schema is shown below:

```
DemoFunction-Single:
  allOf:
    - $ref: 'TS28623_GenericNrm.yaml#/components/schemas/Top'
    - type: object
      properties:
        attributes:
          allOf:
            - $ref: 'TS28623_GenericNrm.yaml#/components/schemas/ManagedFunction-Attr'
            - type: object
              properties:
                demoId:
                  $ref: 'TS28623_GenericNrm.yaml#/components/schemas/DemoId'
                demoFlag:
                  type: boolean
```

```

    exampleConf:
      $ref: '#/components/schemas/ExampleConf'
- $ref: 'TS28623_GenericNrm.yaml#/components/schemas/ManagedFunction-ncO'
- type: object
  properties:
    ChildClass:
      $ref: '#/components/schemas/ChildClass-Multiple'
    AnotherChildClass:
      $ref: '#/components/schemas/AnotherChildClass-Single'

```

Here, a DemoFunction-Single Managed Object Class (MOC) of the solution set is defined corresponding to the *DemoFunction* Information Object Class (IOC) from the information model. A Managed Object Instance (MOI) of a DemoFunction-Single will be a JSON object that conforms to the Top MOC (which defines the id naming attribute) and will also contain an attributes property object. The attributes object conforms to the base class ManagedFunction-Attr and also contains a set of properties defined for the *DemoFunction* including demoId, demoFlag, and exampleConf.

In addition to the attributes object, a DemoFunction-Single conforms to the the ManagedFunction-nc0 MOC and has additional properties corresponding to the name-containment relations defined in the UML diagram for the *DemoFunction* IOC, including multiple *ChildClass* and a single *AnotherChildClass*.

#### 4.1.2.2 NETCONF/YANG Data Models

A fragment of the NETCONF/YANG from the 5G NRM [i.5] is shown below:

```

module demofunction {
...
  grouping DemoFunctionGrp {
    description "Represents the DemoFunction IOC.";
    reference "Example";
    uses mf3gpp:ManagedFunctionGrp;

    leaf demoId {
      type int64 { range "0..68719476735"; }
      mandatory true;
      description "Uniquely identifies the Demo Object at least within a gNB.";
      reference "Example";
    }

    leaf demoFlag {
      type boolean;
      mandatory true;
      description "Turns on something";
      reference "Example";
    }

    list exampleConf {
      key confId;
      config false;
      min-elements 1;
      max-elements 1;
      description "Used to configure something";
      uses ExampleConfGrp;
    }
  }

  augment "/me3gpp:ManagedElement" {

    list DemoFunction {
      key id;
      description "Represents the logical Demo function of gNB or en-gNB.";
      reference "Example";
      uses top3gpp:Top_Grp;
    }
  }
}

```

```

    container attributes {
      uses DemoFunctionGrp;
    }
    uses mf3gpp:ManagedFunctionContainedClasses;
  }
}

```

Here, the attributes from the *DemoFunction* IOC are defined inside a YANG grouping *DemoFunctionGrp*, which in turn is placed inside a YANG container named *attributes* within each element of a list of *DemoFunctions*. The module augments the base *ManagedElement* module to contain such a list, which means that any NETCONF server claiming to implement the *demofunction* may contain a list of *DemoFunctions* inside its *ManagedElement*.

Note that with this approach, the definition of an MOC does not declare its child name-containment relationships, but rather expresses its parent's name containment relations of itself. This is in contrast to the OpenAPI / JSON Schema approach where the child name-containment relations are given in the parent MOC.

### 4.1.3 Extensibility

O1 specifications refer to 3GPP models for the modelling of the logical RAN entities. 3GPP NRM models are defined in 3GPP TS 28.621[i.16] and 28.622[i.6], related YANG solution set is defined in TS 28.623 while YANG mapping rules and vendor extension mechanisms are defined in TS 32.160 [i.9].

Each 3GPP solution set has defined its own set of mapping rules and vendor extension mechanisms to provide a consistent set of models.

Extensibility mechanisms can be used to add non-standardized contents through vendor specific entities in addition to the standardized ones.

In general, the 3GPP Vendor specific extensibility mechanisms can be used by a vendor or by an SDO. In the context of O-RAN Alliance, the O-RAN Alliance community can use vendor specific extensibility mechanisms and define a new standardized data model as O-RAN YANG modules. An O1 producer or consumer to be O1 compliant will have to support new standardized data models.

Vendor extensions to a 3GPP YANG module can be done to:

- 1) Marking in a 3GPP YANG modules as “not supported” any optional element according to supportQualifier and/or any deviations that maintain backward compatibility as defined in RFC 7950

Vendor extensions to a 3GPP YANG module shall be done in a separate YANG module using augmentation mechanism:

- 1) Add vendor specific attributes to a 3GPP specified IOC
- 2) Add a new vendor specific IOC

In case 2 and 3, the vendor extensions are name contained in an existing 3GPP IOC and the new IOC shall follow 3GPP TS 32.160 [i.9] methodology.

3GPP TS 32.160 [i.9] provides examples of case 2) and 3) as follows:

Example 1 – Add a vendor specific attribute to a 3GPP specified IOC:

```

augment /me3gpp:ManagedElement/attributes {
  leaf isCabinetClosed {
    type boolean ;
    description “Indicates whether the doors of the HW cabinet is closed.” ;
  }
}

```

Example 2 – Add a vendor specific IOC:

```

//vendor class
grouping VendorClassGrp {

```

```
// contains all attributes
leaf exampleAttribute {
    type string;
    description vendorMarker;
}

augment /me3gpp:ManagedElement {
    list VendorClass {
        key id;
        uses top3gpp:Top_Grp;
        container attributes {
            uses VendorClassGrp ;
        }
        //YANG lists representing contained classes
    }
}
```

O-RAN has already defined extensions to 3GPP YANG modules, applying 3GPP extensibility principles and has provided O-RAN defined YANG modules. For example, the `o-ran_3gpp-nr-nrm-gnbdufunction.yang` module is an extension of the `3gpp-nr-nrm-gnbdufunction.yang` module where the attributes of the 3GPP `ManagedElement/gnbdu3gpp:GNBDUFunction` IOC have been augmented with additional O-RAN specific attributes (i.e. `mac-configuration-group`) to be used with O-RAN specific namespace (i.e. `o-ran:oran-gnbdufunction`). This is an example of O-RAN Alliance extension of type 2) above.

Exemple 3 – Add vendor specific augment

```
"/me3gpp:ManagedElement/gnbdu3gpp:GNBDUFunction/gnbdu3gpp:attributes" {
    when "(derived-from-or-self(/hw:hardware/hw:component/hw:class, 'o-ran-hw:O-DU-COMPONENT'))";
    container o-ran-configuration {
        uses mac-configuration-group;
        description "Container for o-ran-configuration. These are configured when O-DU is compliant with O1 interface spec for O-DU. ";
    }
}
```

## 4.2 Issues and Recommendations

### 4.2.1 Non-3GPP Information and Data Models

The information modeling methodology and solution sets described in clause 4.1 lead to a very particular and stylized use of YANG and OpenAPI constructs when specifying data models. This arises because of the separation of the attributes of an IOC from the name-contained relationships to other IOCs, and the enforcement of this separation with conventions such as declaring an “attributes” container for each MOC and expressing name-containment with property names (in the case of OpenAPI/YAML) or the “augments” construct (in the case of YANG). (The current O1 specification mandates the use of the YANG Stage 3 solution set rather than the OpenAPI/YAML Stage 3 solution set and so we restrict our discussion to the YANG Stage 3 solution set in what follows).

Non-3GPP SDOs do not necessarily follow the 3GPP methodology of separately modeling the information to be managed at an abstract level and then mapping these models into solution sets. Indeed, many extant YANG specifications from the IETF community are written and edited directly in YANG and make use of the full power of that notation to define data models that, for instance, do not place attributes into a container named “attributes” and which don’t enforce name-containment conventions. The O-RAN Alliance WG4 has developed such a model for the OpenFH M-plane. Finally, vendor-specific YANG models also exist for which no prior information model was developed.

It is not immediately obvious that non-3GPP data models need to be managed via the O1 interface. Perhaps, these models can always be relegated to being managed over other interfaces. For example, the OpenFH M-plane is depicted

as its own interface, separate from O1, and transport elements that are likely to make use of IETF-defined models could be managed solely through a specific interface (not O1) dedicated to that purpose. However, non-3GPP data models may in fact touch the O1 interface in several ways:

1. For the Hierarchical O-RU management model, the O-DU exposes the O-RUs to the SMO via its own O1 interface [i.13]. This has led O-RAN WG5 to propose the embedding of the WG4 OpenFH M-plane data model on a schema mount-point within the 3GPP-compliant O-DU data model [i.14]. Operations that modify the data inside the schema mount cannot refer to MOIs by distinguished name because the OpenFH M-plane is not structured as name-contained MOIs.
2. Although the transport elements themselves may be managed with a separate interface, the O-RAN Network Functions may need to borrow some of the data models to allow for configuration of the connection points where they touch the transport network. Borrowing these models literally from other domains like the IETF instead of defining new 3GPP-compliant ones would lead to simplified operational workflows. Again, these models may not be structured as name-contained MOIs.
3. Any other data models, such as vendor-specific YANGs, that were not developed according to the methodology in 32.156 may also have their own inherent structure which does not follow the name-contained MOI conventions. These YANG models may be embedded in a 3GPP or ORAN-compliant data model, or they may appear as a separate tree rooted at the top level of the management service.

Currently, the O-RAN OAM Architecture [i.13] and O1 interface specification [i.1] define the O1 interface for management of O-RAN Network Functions as a collection of SBMA-compliant Management Services. The existing Stage 2 operations and Stage 3 solution sets defined in 3GPP 28.532 [i.3] assume that the Management Component Type B (NRM) was developed according to the 3GPP methodology outlined in clause 4.1, and therefore cannot be used directly to manage data without distinct attributes and name-contained MOIs. One fundamental reason is that, as outlined in 3GPP TS 32.300 [i.10] clause 4.1, the names of Managed Objects, in the form of Distinguished Names, are passed across various Integration Reference Points (IRPs) and are fundamental to the definition of the operations and cross-references between objects. These limitations will be further outlined in clause 5 of the present document.

#### 4.2.1.1 Issue Summary

Considering data models in general and theoretically, and without declaring any normative requirements in this clause, data to be managed could take any of the following forms, in decreasing order of strictness:

1. Structured data (e.g., JSON or XML data) that conform to a standardized 3GPP Information Model and a YANG data model developed in conformance to the conventions outlined in 3GPP 32.160;
2. Structured data (e.g., JSON or XML data) that conform to a standardized YANG data model that was developed outside of 3GPP conventions, such as in the IETF or O-RAN WG4;
3. Structured data (e.g., JSON or XML data) that conform to a non-standardized YANG data model that was developed outside of 3GPP conventions, such as a vendor-specific model;
4. Structured data (e.g., JSON or XML data) that do not have a YANG data model;

Management of even more loosely structured or unstructured data is also a theoretical possibility but is out of scope for now. Because each of the listed data model sets is a subset of the one that comes after, any management scheme that handles one of the above sets of data models can also handle any of the ones listed before it.

#### 4.2.1.2 Recommendations

We conclude that O-RAN management systems and interfaces should be capable of managing any data of types 1 – 2 from clause 4.2.1.1.

---

## 5 Analysis of Management Operations

### 5.1 Background

#### 5.1.1 MnS Component Type A

Management Service Component Type A is defined in clause 4.2.2 of the 3GPP management and orchestration architecture framework 3GPP TS 28.533 [i.4] as a group of management operations and/or notifications that is agnostic with regard to the entities managed. By “agnostic” it is meant that the operations and/or notifications can be defined

independently of any particular NRM (component type B) because the managed object instances and their data are carried in the payload of the operations (“signature of the operations”, in the words of the architecture framework), but the operations themselves can be applied to many different NRMs.

The 3GPP generic management service specification 3GPP TS 28.532 [i.3], for example, defines a generic provisioning management service. In the stage 2 level of specification in clause 11.1 of 28.532, the operations, notifications, and their parameters are described without regard to how they would be encoded in a stage 3 solution set. Later in clause 12.1, a RESTful HTTP-based solution set is defined that describes how to encode these parameters into an HTTP protocol. In clause 12.1.3, it is specified how these operations and parameters would be encoded in a YANG/Netconf-based solution set.

The stage 2 specification of the generic provisioning management service defines the operations createMOI, getMOIAttributes, modifyMOIAttributes, and deleteMOI, and the notifications notifyMOICreation, notifyMOIDeletion, notifyMOIAttributeValueChanges, notifyEvent, and notifyMOIChanges. For illustrative purposes, we reproduce the input and output parameter tables for the createMOI operation below:

**Table 5.1.1-1: Input parameters of the createMOI operation (from clause 11.1.1.1.2 of 28.532 [i.3])**

| Parameter Name        | S | Information Type / Legal Values                    | Comment                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| managedObjectClass    | M | class                                              | This parameter specifies the class of the new managed object instance.                                                                                                                                                                                                                                                                                                    |
| managedObjectInstance | M | DN                                                 | This parameter specifies the instance of the managed object that is to be created and registered. This is a full DN according to 3GPP TS 32.300 [5].                                                                                                                                                                                                                      |
| attributeListIn       | M | LIST OF SEQUENCE< attribute name, attribute value> | This parameter may have a null value. When this parameter is supplied, it contains a list of name/value pairs specifying attribute identifiers and their values to be assigned to the new managed object. These values override the values for the corresponding attributes derived from the default value set specified in the definition of the managed object's class. |

**Table 5.1.1-2: Output parameters of the createMOI operation (from clause 11.1.1.1.3 of 28.532 [i.3])**

| Parameter name   | S | Matching Information / Legal Values                | Comment                                                                                                                |
|------------------|---|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| attributeListOut | M | LIST OF SEQUENCE< attribute name, attribute value> | This list of name/value pairs contains the attributes of the new managed object and the actual value assigned to each. |
| status           | M | ENUM (OperationSucceeded, OperationFailed)         |                                                                                                                        |

## 5.1.2 Relationship Between Component Type A and Component Type B

As can be seen by inspecting the operation parameters in Table 5.1.2-1 and Table 5.1.2-2, it is clear that while the createMOI operation can be applied to any NRM that was developed according to the 3GPP methodology described in clause 4.1, this particular component type A depends on certain assumptions inherent in the 3GPP design methodology; namely, that managed object instances are instances of a managed object class and can be uniquely identified through their distinguished names. This particular component type A also assumes that the data to be managed is structured as a set of named attributes contained in the managed object instance.



## 5.2 Common Issues and Recommendations

### 5.2.1 Declarative vs. Imperative Pattern

#### 5.2.1.1 Introduction

Declarative and Imperative patterns are different paradigms that describe whether or not to describe the “how”. In operations applying an imperative pattern, the “how” is described explicitly by step-by-step instructions; in operations applying a declarative pattern, the desired state is managed, without explicitly describing about “how” to achieve it.

In the context of this specification, the declarative pattern provides the abstraction to get the element to the desired state. For example, declarative pattern for configuration management involves specifying the desired configuration state rather than explicitly detailing the steps needed to achieve that state.

#### 5.2.1.2 Issue Summary

This clause describes the issues related to the nature of Imperative Pattern of Stage 3 Provisioning operations referred in O-RAN.WG10.O1-Interface [i.1] in 6.1.0 Overview. The clause is not intended to rule out any of the operation requirements of Management Services defined in O1 Specifications, but an effort to pave way towards the usage of Declarative Patterns to achieve those O1 Specified operations. This benefits the consumers to prepare the data and not depend upon the specific implementation of Provisioning operations using YANG/NETCONF solution set.

**Table 5.2.1.2-1: Summary of Issues of Imperative Operating Patterns across various O1 Specified Management Services**

| List of Services (O1 Specified)                                                                                                                                                                                                                                                                                       | Communication Patterns                       | Issue Description with Imperative Pattern of Operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provisioning Management Services <ul style="list-style-type: none"> <li>Create Managed Object Instance</li> <li>Modify Managed Object Instance Attributes</li> <li>Delete Managed Object Instance</li> <li>Read Managed Object Instance Attributes</li> <li>Subscription Control (for modification of MOI)</li> </ul> | "Request-response"<br><br>"Subscribe-notify" | <p>This communication pattern requires Connection-Oriented Operation between MnS Consumer and MnS Producer. The imperative nature of operating patterns associated with Provisioning Management Services reflect in the usage of the imperative primitives offered by the solution set, that incurs upon the MnS consumer to take decisions of ‘when’ and ‘which’ primitive to use to achieve a certain desired state. The issue is that operations of the solution set are imperative in nature and does not give an option for the Producer of Configuration Data (e.g. an rApp) to just output the desired state, i.e., the modified configuration data, that the Consumer of Configuration Data can consume.</p> <p>e.g., an rApp that has to shutdown a certain cell for the purpose of Energy Saving knows the optimization parameter to update and need not bother with the imperative approach and the specific flow of how to make such an update. All that the rApp truly requires is to create the desired state.</p> <p>So, the imperative pattern of operations enforces the consumer to strictly follow the operation methods of the solution set to achieve a certain desired state.</p> |

NOTE: The column ‘Communication Pattern’ refers to clause 5.1.2 Interactions between management service producer and management service consumer of 3GPP TS 28.533 [i.4].

#### 5.2.1.3 Recommendations

We conclude that SMO should use declarative operation patterns to provision the desired state in form of structured data (e.g., JSON or XML data), according to clause 4.2.1.1 of the present document. Consumers of Configuration Data, such as rApp, can create such desired state and it is recommended to apply a declarative pattern through mechanisms of the O-Cloud.

## 5.3 Configuration Management

### 5.3.1 Configuration Repository as Source of Truth

#### 5.3.1.1 Overview

A logically centralized entity or repository hosting the desired configurations for O-RAN NFs intended to be further synchronized to a target NF residing in an O-Cloud environment based on a declarative description. It is referred as logically centralized because the realization of the repository for configuration can be multiple and distributed. Configuration repository is referred as a source of truth in the context of configuration data of NFs and configuration data related to O-Cloud.

Such configuration repository enables access to the configuration data to consumer of configuration data. In the context of provisioning of configuration data, the producer of provisioning management services uses the configuration repository to store the configuration data.

The configuration repository

- enables storing the configuration data from the source of configuration changes [consumer of provisioning MnS],
- hosts the artifacts of configuration (e.g. day 1, day 2, etc.),
- enables delivery of configuration data to the target NFs, based on declarative approach, and
- exposes operations to support create / update / read / delete configuration data.

The consumer of provisioning management services can use methods to push data of desired configuration to the configuration repository. The desired configuration created / modified on the configuration repository, can then be synchronized to the target O-RAN NF(s).

#### 5.3.1.2 Issue Summary

This clause describes the issues related to the nature of operations required between the consumer of provisioning management services and producer of provisioning management services, and issues associated with carrying out the operation procedures defined in stage 3 provisioning operations of 3GPP TS 28.532 [i.3] referred in O-RAN.WG10.O1-Interface [i.1] in clause 6.1.0 Overview.

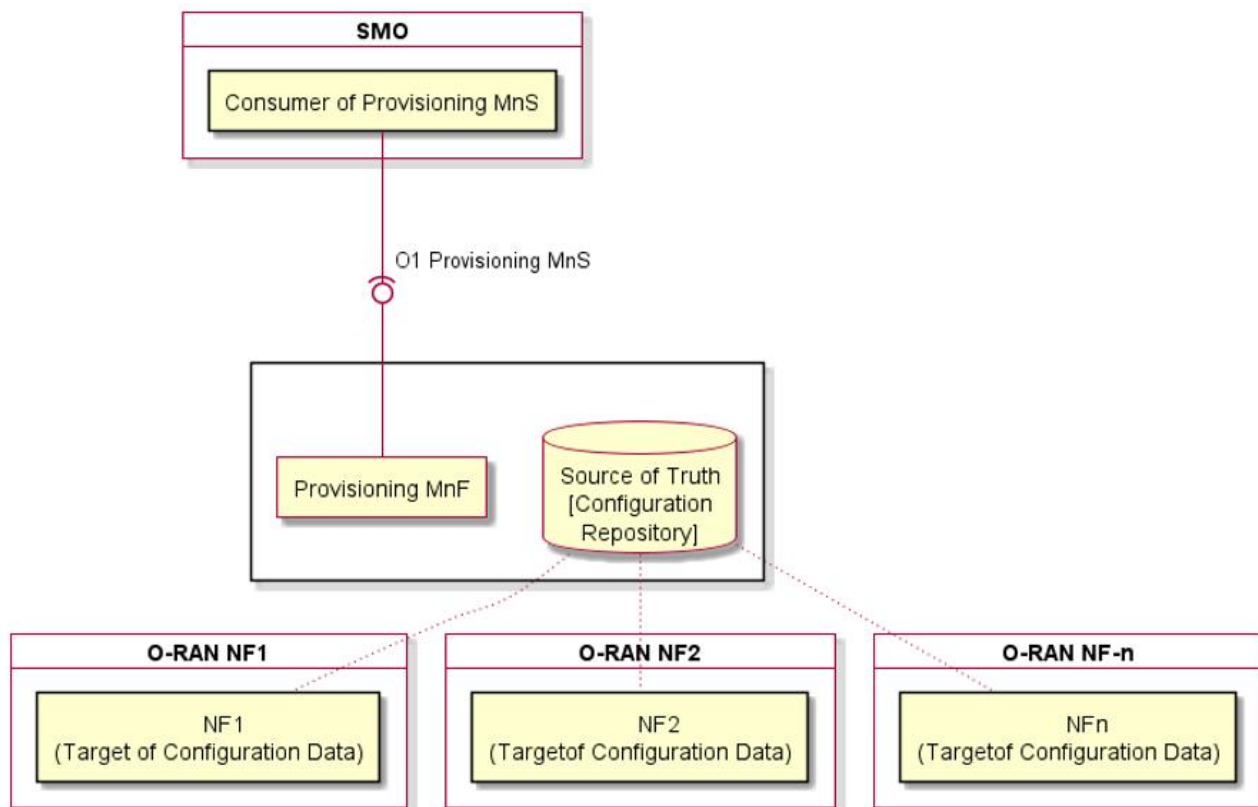
When a consumer of provisioning management services determines to invoke a certain provisioning operation, for instance, to modify or retrieve a configuration parameter on a set of managed object instances (e.g., a number of NR cells data), it has to identify the destination of provisioning MnF peers to send the operation request.

The key issue in such interworking with the provisioning MnF are related to transport of operations and are outlined in clause 5.3.3.2 Issue Summary of the present document.

#### 5.3.1.3 Potential solution

Based on the deployment scenario listed in 3GPP TS 28.533 [i.4], the configuration repository can be placed as shown below in the current OAM architecture to work with the consumer of provisioning MnS. The potential solution for the key issue identified in the above clause (5.3.1.2 Issue Summary) is addressed in the clause 5.3.3.3 potential solution of the present document.





**Figure 5.3.1.3-1: Interconnect Diagram for O1 Provisioning MnS**

## 5.3.2 Configuration Data Changes and Consumers

### 5.3.2.1 Overview

In the O-RAN system, MOIs are representing the management aspects of an O-RAN NF instance. Changes in the MOIs of O-RAN NFs may occur for reasons including, but not limited to, the following:

- 1) Software Updates and Upgrades: O-RAN NFs and rAPPs/xAPPs update and upgrade may result in MOIs value and status change.
- 2) RAN Scaling: New O-RAN NF deployment, scaling, migration or other LCM operations in response to the capacity optimization requirement, the O-RAN NF LCM may trigger configuration changes reflected in the O-RAN NF MOI, e.g., vnfParametersList modifications as defined in [i.6].
- 3) Fault Recovery: In the event of a NF fault, configuration changes may be needed to isolate problematic NFs, or enable fault recovery.
- 4) Security Updates: The encryption settings as one kind of configuration data will be updated with the development of new integrity and encryption algorithms.
- 5) Integration of New Technologies/Features: The integration of new technologies, such as the introduction of new spectrum bands, Carrier Aggregation (CA), sidelink, network slice, etc., usually requires configuration changes to the corresponding features.
- 6) Automated RAN Management: With the introduction of AI/ML in RAN, networks are progressively becoming automated. In O-RAN, configuration changes can be triggered by CI/CD pipelines, Non-RT RIC for RAN data collection, automated configuration/optimization (e.g., C-SON, D-SON), as well as triggered by the deployment and removal of xAPPs.

The sources of configuration data change summarized above will result in the MOI creation, deletion, and updating of MOI attributes. Changes in MOIs are reported with the Notify Managed Object Instance Changes message via the O1

interface to the RAN NF OAM SMOSs producer. Furthermore, other SMOS producers may access modified configuration data by consuming the RAN NF OAM SMOSs.

From the monolithic view of the SMO, as per [i.13] defined use cases and O-RAN OAM architecture, the SMO is the MnS consumer who can consume the MnS provided by O-RAN NFs via O1 interface.

Meanwhile, under the SBA design concept, the Decoupled SMO architecture [i.31] provides a logical architecture for a decoupled SMO as depicted in clause 6.2.1 in [i.31]. In the decoupled SMO architecture, the terminology of SMO Service (SMOS) and SMO Function (SMOF) are defined, and SMOSs can be consumed by SMOFs via the SMO communication. In the decoupled SMO, RAN NF OAM SMOSs' producer consume the O1 MnS and offer to other RAN NF OAM SMOS consumer RAN NF OAM SMOSs such as RAN NF Provisioning Management SMO Service (NFCM), RAN NF Performance Assurance Management SMO Service (NPFM), and RAN NF Fault Supervision Management SMO Service (NFFM) to other SMOFs. The Decoupled SMO Architecture clause 5.1 in [i.31] illustrates an example based on [i.23], where an rApp consumes RAN NF OAM SMOSs to collect OAM data for the purpose of analysis and the generation of resource allocation/optimization actions to ensure the quality of service for UEs. It also highlights that different intelligent use cases require different types of RAN management data, including performance data, alarm data, and NF NRM-related data.

In this scenario, RAN NF OAM SMOSs have multiple delivery methods for consumption by other SMOFs. For instance, RAN NF OAM SMOSs can be directly exposed to other SMOFs as MnS in order to consume MnS components A, B and C produced by O-RAN NFs.

The table below provides some examples of consumers of OAM management data inside SMO.

**Table 5.3.2.1-1: Examples of consumers of OAM management data inside SMO**

| Potential consumers                                | Potential consumed data | Description                                                                                                                                                                                                                   |
|----------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rApp                                               | MnS component A, B, C   | The configuration data can be utilized for intelligent use cases defined in [i.23]                                                                                                                                            |
| Service & Subnet Slice Orchestration SMOS producer | MnS component A, B, C   | The configuration data is consumable to Network Slice Subnet Management (NSSM) use cases as defined in [i.24]. A specific example of Radio Resource Management (RRM) Policy attributes is delineated in clause 7.8 in [i.24]. |

### 5.3.2.2 Issue Summary

The OAM SMOSs may register with the SME. Each OAM SMOS such as NF CM, NF PM, NF FM, etc. provides its own API for registration with the SME. Consumers of OAM SMOSs may retrieve the individual API of each OAM SMOSs through the service discovery and consume the OAM SMOS with the retrieved API.

NOTE: The process by which OAM SMOSs producer register services to SMEs, in other words, the method for service registration, will adhere to the relevant specification.

### 5.3.2.3 Potential solution

The RAN NF OAM SMOSs APIs, schema of data needs to be defined.

## 5.3.3 Transport of Configuration Management Operations

### 5.3.3.1 Overview

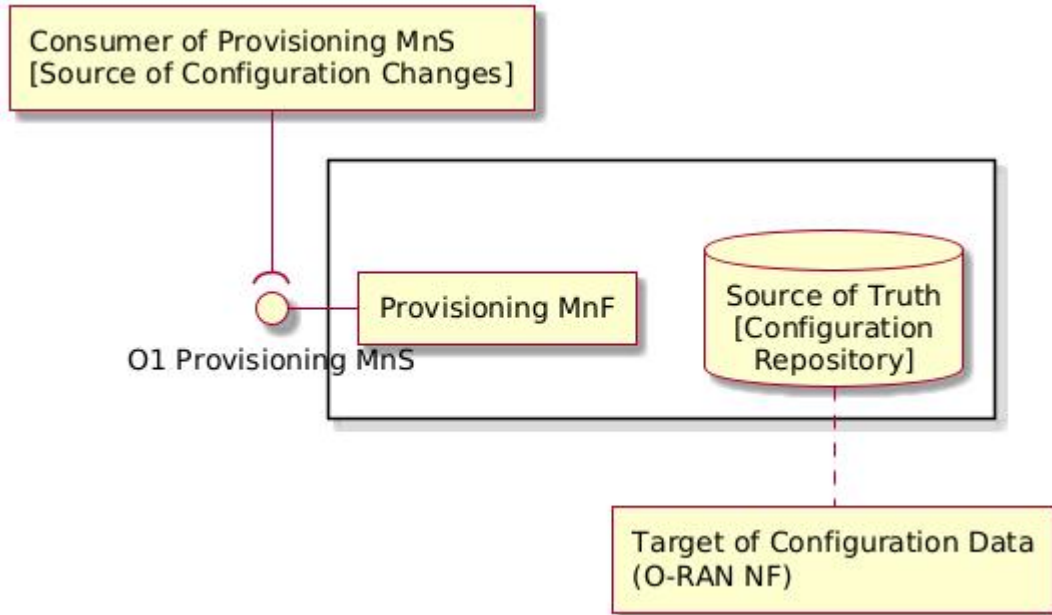
The transport of configuration management operations specified in O1 specifications make use of NETCONF over SSH or NETCONF over TLS.

This clause focuses on how the consumer of provisioning MnS, i.e., the source of configuration changes, can invoke management operations on the configuration data.

Management of configuration data encompasses the creation, modification, deletion of configuration data by the management systems, e.g., SMO, via operations provided by the provisioning management service which uses the

configuration repository to store the configuration data. The transport of configuration management between the consumer of provisioning MnS and the source of truth is agnostic to the transport / protocol support on the target of configuration data.

Figure 5.3.3.1-1 illustrates the configuration repository in the context of the provisioning management service



**Figure 5.3.3.1-1: Provisioning MnS in the context of source of truth**

For instance, SMO / Non-RT RIC acting as consumer of the provisioning MnS, can modify the configuration data for an MOI, and use the operations supported by the configuration repository, to persist the data towards the source of truth. Provisioning MnF contains notification capabilities that may be used by the source of configuration changes [consumer of provisioning MnS] to get the updates about the changes on O-RAN NFs.

### 5.3.3.2 Issue Summary

This clause describes the issues related to the nature of transport of the operations used between the consumer of provisioning management services (i.e., SMO) and producer of provisioning management services (i.e., O-RAN NF) of Stage 3 provisioning operations referred in O-RAN.WG10.O1-Interface [i.1] in clause 6.1.0 Overview. As explained in O-RAN.WG10.O1-Interface [i.1], provisioning management services allow a provisioning MnS consumer to configure attributes of managed objects (configuration management) on the provisioning MnS producer that modify the provisioning MnS producer's configuration in its role in end-to-end network services and allows a provisioning MnS producer to report configuration changes to the provisioning MnS consumer.

The O1 interface specification defines usage of the following solution set:

- YANG/Netconf-based solution set

Netconf (Network Configuration) protocol is a standard interface for installing, deleting and modifying the configuration data on network devices. The issue with Netconf protocol is that there must be a permanent session (SSH / TLS session) to make the communication successful. The RESTful HTTP is the common communication protocol for cloud-based solutions and used for integrating with cloud platform communication patterns. The RESTful HTTP-based protocol as defined in 3GPP TS 28.532 [i.3] in clause "12.1.1 RESTful HTTP-based solution set" is not currently specified for the O1 interface.

RESTful HTTP communication protocol for cloud-based solution supports scaling across distributed cloud environments, minimizes overhead (light-weight), tolerates failure by leveraging self-healing mechanisms, and supports asynchronous communication. Adapters are required to support the existing provisioning management service solution set (YANG/Netconf) to terminate to a cloud based RESTful HTTP solution.

A consumer of provisioning management service that makes use of the solution set listed above (YANG/Netconf-based solution set), requires a change in the transport mechanism for supporting the cloud friendly RESTful HTTP-based protocol for configuration management operations, to make use of commands or APIs for management of configuration data while interworking with the provisioning MnF.

### 5.3.3.3 Potential solution

Based on the gaps in O1 interface compared to the solution sets provided in 3GPP, one potential solution is to enhance the O1 interface specification with the alternative solution set (JSON/HTTP) from 3GPP for provisioning management service as defined in 3GPP TS 28.532 [i.3] in clause “12.1.1 RESTful HTTP-based solution set”.

## 6 Analysis of Operations for Telemetry Control

### 6.1 Background

#### 6.1.1 MnS Component Type C

MnS component type C, as defined in 3GPP TS 28.533, represents the performance and fault information of managed entities. It provides essential data for monitoring and maintaining the operational health of network elements, facilitating effective performance management and fault detection. MnS component type C exposes management data to RAN NF OAM SMOS. Performance information, fault information are network management data collected from the network (including e.g. service, slicing and/or network functions related data). RAN analytics SMOs can consume RAN NF OAM SMOs to utilize the network management data and make the corresponding analytics based on the collected information. The performance information provided by RAN NF MnS producer can be used to optimize network performance, and the fault information provided by RAN NF MnS producer can be used to predict and prevent failures of the network.

Mobile networks have the capability to support a wide variety of services and requirements. This, along with increasing flexibility of the network may present management and operational challenges and complexities. The management system can therefore benefit from management data analytics services for improving networks performance and efficiency to accommodate and support the diversity of services and requirements.

MnS component type C consists of below primary elements:

- Performance Data is defined in [i.28], 3GPP TS 28.552 and TS 28.554.
- Alarm Information is described in 3GPP TS 28.532 and TS 28.111.

NOTE: The alarm information also includes alarms defined in O-RAN WG10 WI O-RAN-WG10.O1.Alarms, which will be referenced here once the O-RAN-WG10.O1.Alarms specifications are published.

- Traces information as defined in TS 32.421 and TS 32.423.

Trace information is diverse, and it mainly supports troubleshooting and validation testing. The Trace records should contain Information Elements or signaling messages from control signaling and/or the characteristics of the user data.

Together with MnS component type A and MnS component type B, MnS component type C forms a comprehensive management service that supports the configuration, monitoring, and optimization of network resources.

In the O-RAN architecture, MnS component type C can be accessed through the O1 interface. The SMO utilizes the O1 interface to collect and process MnS component type C information, enabling the monitoring and management of the performance and fault status of O-RAN NFs. MnS component type C can also be provided to the Non-Real-Time RIC through the RAN NF OAM SMOS producer in the SMO, enabling intelligent network performance optimization, fault detection, and recovery.

#### 6.1.2 Performance Metrics

3GPP TS 32.401 defines the concept of Performance Management activity as one to collect data, which can be used to verify the physical, virtual, and logical configuration of the network and to locate potential problems as early as possible. The type of data to be collected is defined by the equivalent measurements (e.g., as specified in 3GPP TS 32.

425 for E-UTRAN related performance measurements, and 3GPP TS 28.552 for NR related performance measurements). O-RAN O1 Interface Specification refers to 3GPP 28.622 for the performance management job operation, based on the PerfMetricJob IOC, where a MnS consumer lists the performance measurements to be produced (performance metrics are identified by name). All object instances below and including the instance name-containing the PerfMetricJob are scoped for performance metric production.

In measurement data reporting, according to 3GPP 32.404, a performance measurement includes the measurement object class, describing the measured object class level where the measurement has been collected (e.g. NRCellCU, NRCellRelation).

Performance measurements, which are a permanent source of information, provide values aggregated on an observation period.

### 6.1.3 Events and Alarms

Event is defined as network occurrences which have significance for the management of an NE (Network Element) in clause 3.1 Terms of 3GPP TS 28.111 [i.21], for example a configuration change, a threshold crossing, a transition to an error state or a transition to a failure state. Events do not have state.

3GPP TS 28.111 [i.21] defines in clause 3.1 Terms, an alarm as a representation of an error or failure that requires attention or reaction by an operator or some machine. Alarms have state.

### 6.1.4 Traces

Introduction of 3GPP TS 32.421 [i.19] cites that Subscriber and equipment trace provide very detailed information at call level on one or more specific mobile(s). Such data is an additional source of information to Performance Measurements and allows going further in monitoring and optimization operations. Subscriber and Equipment (UE) Trace give instantaneous values for a specific event (e.g. call, location update, etc.). Subscriber tracing includes tracing of IMSI or Public User Identity, and equipment (UE) tracing includes tracing of IMEI or IMEISV. Trace is the easy way to go deeper into investigation and network optimization.

The use cases for Trace can be divided into two basic categories:

- Troubleshooting use cases cover situations where the operator is solving an existing problem in the operator network.
- Validation testing use cases cover situations where the operator is not solving a known problem but merely analyzing, fine-tuning or optimizing the network.

Subscriber and UE Trace are commonly used for the purposes (to list a few) such as:

- Checking radio coverage in certain areas, testing radio coverage on a particular geographical area following network extension for instance (e.g. new site installation) by collecting RLF reports,
- Fine-tuning and optimization of algorithms/procedures; Trace data are used to get feedback on the network quality and capacity after optimization operations like parameter fine-tuning, for instance, by using measurement data (Measurement Report).
- Periodic sampling of network performance.

---

## 7 Analysis of Transport of Telemetry

### 7.1 Background

#### 7.1.1 Transport of Performance Metrics in Current SBMA Solution Sets

The O1 Interface [i.1] specifies two ways to transport Performance Metrics:

1. File-based
2. Streaming

The following focuses on describing Performance Data streaming as it is expected to be more relevant to cloud-native OAM.

Performance data streaming makes use of WebSockets as a transport for the performance data [i.1, i.3]. The consumer of the O1 PM Service in the SMO receives the PM data over the WebSockets connection and makes it available to other consumers within the SMO (e.g., the Non-RT RIC and its rApps).

The performance data streaming MnS consists of two related protocol interfaces:

- HTTP / HTTPS for the following purposes:
  - Establishment of data pipe for streaming (typically, the creation of WebSocket connection). The stream establishment operation contains the Producer ID as a parameter; if the MnS Producer is modeled as part of the NRM, then its distinguished name (DN) can be used here. Otherwise, the Producer ID is filled in with some other unspecified identifier.
  - Exchange of meta-data (the nature of the data to be reported via streaming). This enables the Producer to inform the Consumer of the format and sequence of the data streamed on multiple streams.
  - Management of reporting streams based on the stage 3 definition of “establishStreamingConnection”, “addStream”, and/or “deleteStream” operations.
- WebSockets as a transport channel for Performance Data:
  - Formatted and packed as performance data payloads as per Annex G or Annex I of TS 28.550v18 (either in ASN.1 packet or GPB packet) [i.15], and

Sent over the WebSocket binary data frames (based on the stage 3 definition of "reportStreamData" operation) [i.3]. The delivery of WebSocket Data frame is taken care of by the underlying TCP.As an example, it is shown below how two streams identified by streamId 123 and streamId 124 of performance data are characterized using stream information meta-data conveyed by the Producer to the Consumer using an "establishStreamingConnection" operation.

POST /example.com/o1\_pm\_trace\_streaming/1.0.0/connections

accept: application/json

Content-Type: application/json

```
{
  producer: "SubNetwork=ABCNetwork,SubNetwork=MUC01,GNBDUFunction=XYZ0100",
  streams: [
    {
      streamType: "PERFORMANCE",
      serializationFormat: "GPB",
      streamId: "123",
      additionalInfo: {
        measObjDn: {
          GNB-ID: "1",
          NCGI: "2"
        }
      },
      performanceMetrics: [
        "MR.RSRP.00",
        "MR.RSRP.01",
        ...
      ],
      jobId: "Job-Id-x"
    },
    {
      streamType: "PERFORMANCE",
      serializationFormat: "GPB",
      streamId: "124",
      additionalInfo: {
        measObjDn: {
          GNB-ID: "1",
          NCGI: "2"
        }
      }
    }
  ]
}
```



```

    }
  },
  performanceMetrics: [
    "ERAB.EstabInitAttNbr.1",
    "ERAB.EstabInitAttNbr.6",
    "ERAB.EstabInitAttNbr.9"
  ],
  jobId: "Job-Id-x"
}
]
}

```

The example shows request for reporting data streams meta-data for two streams which contain separate sets of Performance Metrics data belonging to different measurement families. Alternately, separate streams of reporting data may be conveyed for performance data generated by different measurement objects, identified distinctly by “measObjDn”. Once the meta-data of reporting stream is successfully acknowledged by the Consumer of Consumer of MnS for Performance Data Streaming, the Consumer shall use this meta-data to correlate with the payload received on WebSocket data frame, identified based on the stream identifier information, to apply decoding, unpacking, specifically to extract and read the required Performance Metrics. That is, the sending of Performance Metrics data over GPB or ASN.1 format must abide by the Performance Metrics’ list that was exchanged as part of meta-data between Producer of MnS and Consumer of MnS. This is to ensure on the Consumer that a certain PM is expected only at a position specified by the Performance Metrics (*performanceMetrics* field of “establishStreamingConnection” operation). The corresponding payload carrying the actual measurement results is packed in the GPB / ASN.1 encoded message fields, serialized and sent using WebSocket's data frame. A representation of how the data looks along with the interpretations of the message fields for payload for streams in our example are shown below.

#### Reporting of Performance Data for streamId: 123

| GPB Structure/Fields        | Values received in Payload and interpretations by Consumer             |
|-----------------------------|------------------------------------------------------------------------|
| PDSUs                       |                                                                        |
| PDSU                        |                                                                        |
| stream_id                   | 123                                                                    |
| granularity_period_end_time | 2022-02-02T19:34:00.000Z                                               |
| standardized_meas_results   |                                                                        |
| real_value                  | 50.0 [received/interpreted by Consumer of PMs as value for MR.RSRP.00] |
| standardized_meas_results   |                                                                        |
| real_value                  | 34.0 [received/interpreted by Consumer of PMs as value for MR.RSRP.01] |
| standardized_meas_results   |                                                                        |
| real_value                  | ...                                                                    |

#### Reporting of Performance Data for streamId: 124

| GPB Structure/Fields        | Values received in Payload and interpretations by Consumer                                                   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------|
| PDSUs                       |                                                                                                              |
| PDSU                        |                                                                                                              |
| stream_id                   | 124                                                                                                          |
| granularity_period_end_time | 2022-02-02T19:34:00.000Z                                                                                     |
| standardized_meas_results   |                                                                                                              |
| real_value                  | 6.0 à received/interpreted by Consumer of PMs as value for ERAB.EstabInitAttNbr.1 (QCI: 1 level sub-counter) |
| standardized_meas_results   |                                                                                                              |
| real_value                  | 8.0 à received/interpreted by Consumer of PMs as value                                                       |

|                           |                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------|
|                           | for ERAB.EstablnitAttNbr.6 (QCI: 6 level sub-counter)                                                         |
| standardized_meas_results |                                                                                                               |
| real_value                | 10.0 à received/interpreted by Consumer of PMs as value for ERAB.EstablnitAttNbr.9 (QCI: 9 level sub-counter) |

Above is a representation of Performance Data produced by the Producer of Performance Data Streaming MnS that is encoded in GPB or ASN.1 encoding format and sent over a WebSocket Data Frame towards the Consumer of Performance Data Streaming MnS.

## 7.1.2 Transport of Events and Alarms in Current SBMA Solution Sets

### 7.1.2.1 Transport of Events

According to 3GPP TS 28.111 [i.21], event is defined as anything that occurs at a certain point in time, for example a configuration change, a threshold crossing, a transition to an error state or a transition to a failure state. Events do not have states.

The Event Notifications are configured using Subscription Control, that allows a Provisioning MnS Consumer to subscribe to notifications emitted by a Producer of Provisioning MnS.

The Producer of Provisioning MnS sends notifyMOIChanges notification to the Provisioning MnS Consumer. The O1-supported 3GPP-specified CM notification is:

- notifyMOIChanges

As per O1 specifications, a single notifyMOIChanges notification can report one or more MOI creations, MOI deletions and/or MOI attribute value changes in one notification.

An event notification uses either SDO O1 Format or VES O1 Format to transport the O1 CM notification data to the Consumer of Provisioning MnS, as a feedback loop, using the following:

### 7.1.2.2 Transport of Alarms

According to 3GPP TS 32.111-1 [i.26], an alarm signifies an undesired condition of a resource (e.g., network element, link) for which an operator action is required.

A deviation of a system from normal operation, which may result in the loss of operational capabilities of the element or the loss of redundancy in case of a redundant configuration is referred to as a 'fault'.

The Fault Notifications are configured using Subscription Control, that allows a Fault Supervision MnS Consumer to subscribe to notifications emitted by a Fault Supervision MnS Producer.

Fault Supervision Management Service allows transport of alarms via notification that are listed below:

- notifyNewAlarm
- notifyChangedAlarmGeneral and/or notifyChangedAlarm
- notifyClearedAlarm
- notifyAlarmListRebuilt

According to clause 6.2 Fault Supervision Management Services of O-RAN.WG10.O1-Interface [i.1], the transport of faults use the following services:

- Fault supervision notifications that allow the Producer of Fault Supervision MnS to send alarm notifications, which includes subscription to alarms, notification of new alarms or sending changed alarm notification,
- Fault supervision that allows the Consumer of Fault Supervision MnS to get alarm list from the Producer of Fault Supervision MnS.



Both the services, use either SDO O1 Format, or VES O1 Format to transport the O1 alarm data to the Consumer of Fault Supervision MnS, using following solution set as specified in 3GPP TS 28.532 [i.3]:

- RESTful HTTP-based solution set,
- RESTful HTTP-based solution set for integration with ONAP VES API

Both the solution sets make use of secured HTTP transport to carry the alarm information either as operation response or in form of notifications. The solution set associated with operations and notifications of both the Fault Notifications, and the Fault Supervision Control require connection managements, at HTTP/TLS protocol level which is used for point-to-point communication between each source and target peer of Fault Supervision MnS Consumer and Fault Supervision MnS Producer.

### 7.1.3 Transport of Traces in Current SBMA Solution Sets

The O1 Interface [i.1] specifies two ways to transport traces:

1. File-based
2. Streaming

This clause focuses on describing transport of trace data based on streaming.

Trace data streaming makes use of web sockets as a transport for the trace data [i.1, i.3]. The consumer of the O1 trace service in the SMO receives the trace data over the web sockets connection and makes it available to other consumers within the SMO (e.g., Non-RT RIC, rApps, etc).

The trace data streaming MnS consists of two related protocol interfaces:

- HTTP / HTTPS for the following purposes:
  - Establishment of data pipe for streaming (typically, the creation of web socket connection). The stream establishment operation contains the producer ID as a parameter; if the MnS Producer is modeled as part of the NRM, then its distinguished name (DN) can be used. Otherwise, the producer ID is filled in with some other unspecified identifier.
  - Exchange of meta-data (the nature of the data to be reported via streaming). This enables the producer to inform the consumer of the format and sequence of the data streamed on multiple streams.
  - Management of reporting streams based on the stage 3 definition of “establishStreamingConnection”, “addStream”, and / or “deleteStream” operations.
- Web sockets as a transport channel for trace data:
  - Formatted and packed as trace data payloads as per Annex G.2 of TS 32.423 v18 (GPB packet) [i.27], and sent over the web socket binary data frames (based on the stage 3 definition of “reportStreamData” operation) [i.3]. The delivery of web socket data frame is taken care of by the underlying TCP.

As an example, it is shown below how stream identified by streamId “4358070034D7” of trace data is characterized using stream information meta-data conveyed by the producer to the consumer using an “establishStreamingConnection” operation.

POST /example.com/o1\_pm\_trace\_streaming/1.0.0/connections

accept: application/json

Content-Type: application/json

```
{
  "producer": "SubNetwork=ABCNetwork,SubNetwork=MUC01,GNBDUFunction=XYZ0100",
  "streams": [
```

```

{
  "streamType": "TRACE",
  "serializationFormat": "GPB",
  "streamId": "4358070034D7",
  "additionalInfo": {
    "jobType": "TRACE_ONLY",
    "traceReference": {
      "mcc": "435",
      "mnc": "807",
      "traceId": "0034D7"
    },
    "jobId": "Job-Id-x",
    "traceReportingFormat": "STREAMING",
    "traceTarget": {
      "TargetIdType": "gNB",
      "TargetIdValue": "1"
    },
    "traceConfig": {
      "listOfInterfaces": {
        "gNB-CU-CPInterfaces": [
          "Uu",
          "X2-C"
        ]
      },
      "listOfNeTypes": "GNB_CU_CP"
    }
  }
}
]
}

```

Below is shown an example of a trace data record corresponding to the above connection established.

record

```

{
  header
  {

```

```

time_stamp:1706606407
nf_instance_id:"NETWORK_MANAGED_ELEMENT_ID"
nf_type:"gNB"
trace_reference:"0034D7"
global_gnb_id:
{
  plmn_identity:"435807"
  gnb_id:1780
}
}
payload:
{
  payload_size:0
  binary_payload:"\n\x01\t\x11"
}
}

```

## 7.2 Performance Metric Transport

### 7.2.1 Overview

The transport of performance metric specified in O1 specifications is file-based (bulk) and/or streaming (real time).

For file-based (bulk) performance data reporting, PM data pull-based file reporting method and/or the PM data push-based file reporting method is supported. Asynchronous notifyFileReady notification event is supported when PM File(s) is ready for retrieval or asynchronous notifyFilePreparationError notification event is supported when there is any error while preparing the file(s), which is optional for push-based file reporting. When transferring files, a secure connection is established from or to the File Management MnS Producer (SFTP, FTPeS or HTTPs) as defined in clause 7.1.3 of 3GPP TS 28.537 [i.29].

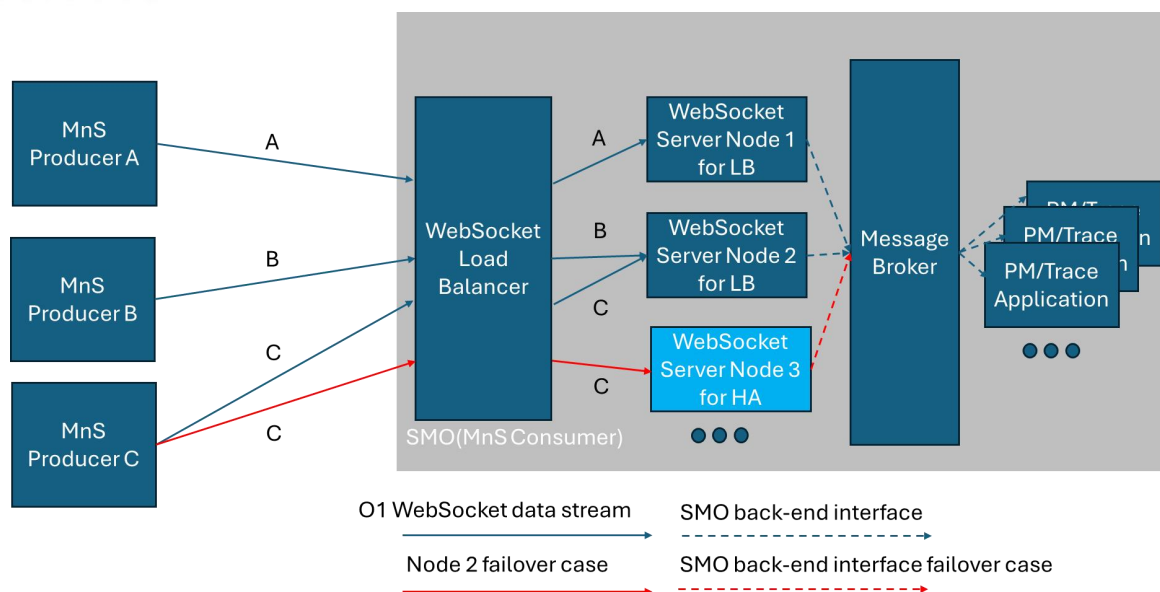
Performance data file content and file naming is as specified in clause 11.3.2.1 of 3GPP TS 28.532 [i.3].

Performance data file XML Format is as specified in clause 12.3.2 of 3GPP TS 28.532 [i.3] and/or in clause 4.1 of 3GPP TS 32.432 [i.30].

For performance data streaming, A secure WebSocket connection is established between the Performance Assurance MnS Producer and the Performance Assurance MnS Consumer. The description of performance data streaming is in clause 7.1.2 Transport of Performance Metrics in Current SBMA Solution Sets of the present document.

### 7.2.2 Issue summary

Figure 7.2.2-1 provides an example of PM data streaming based on WebSocket architecture. During failover scenario, MnS producer need to reestablish WebSocket connection to MnS consumer as indicated by red arrow in Figure 7.2.2-1.



**Figure 7.2.2-1 An Example of PM data streaming based on WebSocket**

WebSocket is point-to-point persistent connection. To maintain it, streaming data reporting MnS producer and consumer need keep alive messages, that costs extra resource.

WebSocket is stateful connection in nature. Streaming data reporting MnS producer and consumer need to maintain state or consider stateless solution in cloud-native deployments, which introduces architectural and operational complexities.

WebSocket doesn't support auto-reconnect by default. WebSocket reconnection time mainly depends on failure detection time and reconnection strategy according to configuration. WebSocket reconnection interrupt time may impact PM data streaming transmission.

WebSocket doesn't support message queueing. To ensure delivery guarantees, Streaming data reporting MnS producer and consumer need additional processing.

### 7.2.3 Potential solution

Compared with WebSocket, Kafka is a message bus (not point-to-point) and stateless, so MnS producer and consumer don't need to maintain peer state. Other issues described in clause 7.2.2 are also handled by Kafka which is described in clause 7.3.3.

## 7.3 Notification Transport

### 7.3.1 Overview

REST/HTTPS is specified as transport of O1 Notification in [i.1]. This transport is based on point-to-point connection between Notification reporting service MnS producer and MnS consumer.

### 7.3.2 Issue summary

Below diagram shows an example architecture based on REST/HTTPS protocol.

```
@startuml
left to right direction
frame SMO #grey;line:grey;line.bold;text:white{
    rectangle LB as "HTTP\nLoad\nBalancer" #steelblue;text:white
    rectangle SN3 as "HTTP Server\nNode 3 for\nHigh Availability" #deepskyblue;text:white
    rectangle SN2 as "HTTP Server\nNode 2 for\nLoad Balancen" #steelblue;text:white
    rectangle SN1 as "HTTP Server\nNode 1 for\nLoad Balancen" #steelblue;text:white
    rectangle MB as "Message\nBroker" #steelblue;text:white
```

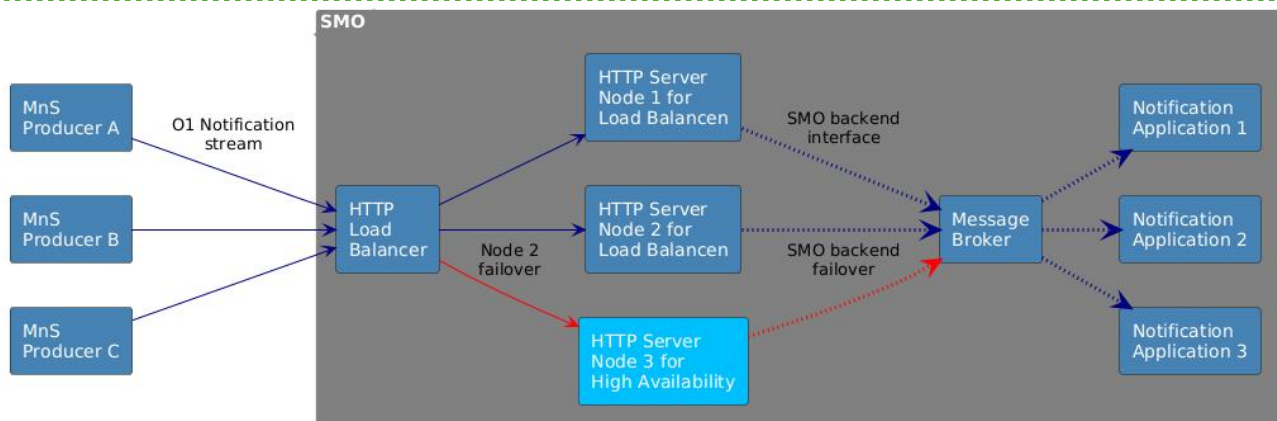
```

rectangle NA3 as "Notification\nApplication 3" #steelblue;text:white
rectangle NA2 as "Notification\nApplication 2" #steelblue;text:white
rectangle NA1 as "Notification\nApplication 1" #steelblue;text:white
LB -[#red]-> SN3 : Node 2\nfailover
LB -[#navy]-> SN2
LB -[#navy]-> SN1
SN3 -[#red,dotted,thickness=3]-> MB : SMO backend\nfailover
SN2 -[#navy,dotted,thickness=3]-> MB
SN1 -[#navy,dotted,thickness=3]-> MB : SMO backend\ninterface
MB -[#navy,dotted,thickness=3]-> NA3
MB -[#navy,dotted,thickness=3]-> NA2
MB -[#navy,dotted,thickness=3]-> NA1
}

rectangle PA as "MnS\nProducer A" #steelblue;text:white
rectangle PB as "MnS\nProducer B" #steelblue;text:white
rectangle PC as "MnS\nProducer C" #steelblue;text:white

PA -[#navy]-> LB : O1 Notification\nstream
PB -[#navy]-> LB
PC -[#navy]-> LB
@enduml

```



**Figure 7.3.2-1 An example architecture based on REST/HTTPS protocol**

In Cloud-native deployment, REST/HTTPS can support load balancing and scalability capabilities well. But there're following issues:

1. There's no guarantee of in-sequence-delivery for HTTP POST operation.
2. HTTP protocol header overhead (including HTTP header, Request/Response mechanism and TCP head-of-line blocking) increases network bandwidth usage especially in large scale Cloud-native deployment.
3. REST/HTTPS protocol doesn't involve any load balancing related aspect. To support load balancing, need to introduce a dedicated load balancer such as: Reverse Proxy which requires additional design complexity and implementation effort.

### 7.3.3 Potential Solution

Below diagram shows an example architecture based on Kafka protocol:

```

@startuml
left to right direction
frame SMO #grey;line:grey;line.bold;text:white{
frame "Kafka partition 1" #lightsteelblue;line:grey;line.bold;text:white{
rectangle KF1 as "Kafka\nReplica\n(follower)" #steelblue;text:white
rectangle KL1 as "Kafka\nReplica\n(lead)" #steelblue;text:white
}
rectangle KP2 as "Kafka partition 2" #lightsteelblue;line:grey;text:white
rectangle NA3 as "Notification\nApplication 3" #steelblue;text:white
rectangle NA2 as "Notification\nApplication 2" #steelblue;text:white
rectangle NA1 as "Notification\nApplication 1" #steelblue;text:white
}

```

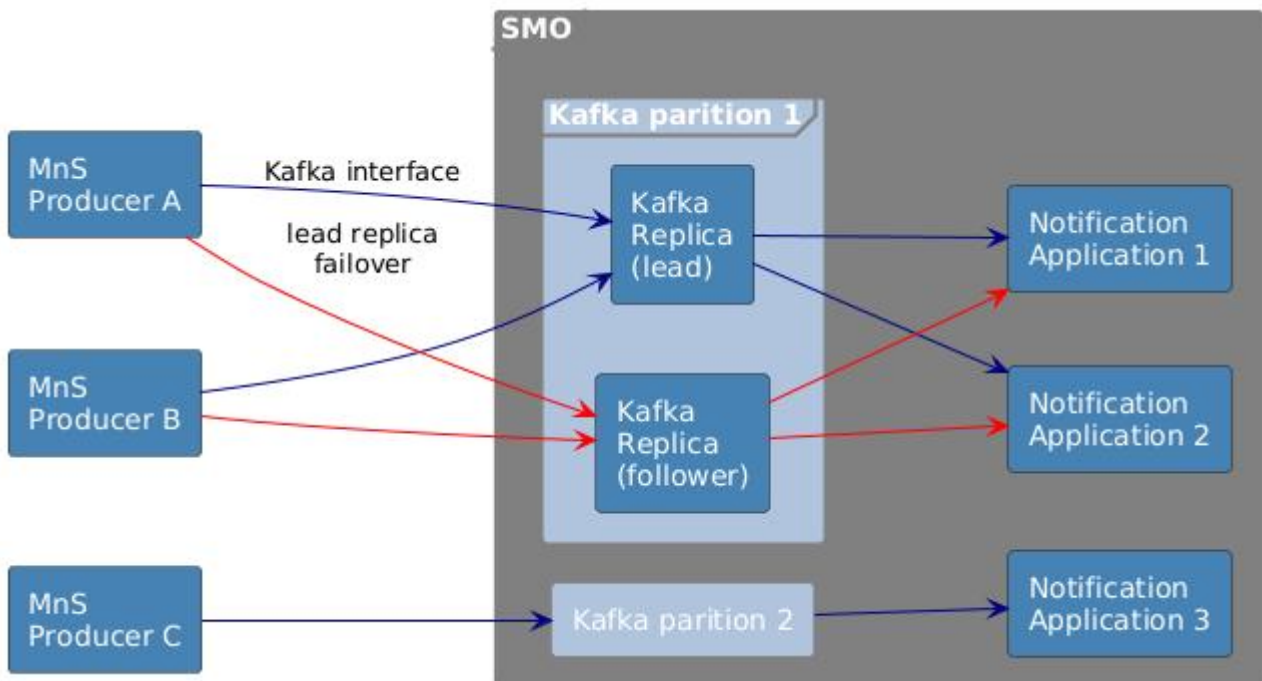
```

}

rectangle PA as "MnS\nProducer A" #steelblue;text:white
rectangle PB as "MnS\nProducer B" #steelblue;text:white
rectangle PC as "MnS\nProducer C" #steelblue;text:white

PA -[#navy]-> KL1 : Kafka interface
PB -[#navy]-> KL1
PA -[#red]-> KF1 : lead replica\nfailover
PB -[#red]-> KF1
PC -[#navy]-> KP2
KL1 -[#navy]-> NA1
KL1 -[#navy]-> NA2
KF1 -[#red]-> NA1
KF1 -[#red]-> NA2
KP2 -[#navy]-> NA3
@enduml

```



**Figure 7.3.3-1 An example architecture based on Kafka protocol**

The issues described in clause 7.3.2 are handled by Kafka as below:

1. In-sequence-delivery is guaranteed in each partition based on in sequence storage.
2. Kafka provides a low protocol header overhead ( multiple messages in one batch and share a common message header). Kafka is still based on TCP, but TCP head-of-line blocking issue is resolved by multiple partitions and sending messages in batch.
3. Kafka protocol includes group id which guarantees multiple consumers in one group have load balancing and fail-over capability. In addition, Kafka heart-beat interval is short (second level), so failure detection is soon.
4. Kafka is not only a protocol, but also a product which handles all aspects of load balance, fault-tolerance and data integrity. It enables application layer to push these features to transport layer.

Based on the above advantages, Kafka could be considered as a potential solution. In case Kafka is introduced as additional transport protocol for O1 interface, following options are available for consideration:

**Option 1** – MnS consumer (SMO) support all solution sets, MnS producer (NF) side support any of the solution set

**Option 2** - Both MnS consumer (SMO) and producer (NF) support any of the options, interoperability is left to operator/vendor product order negotiation

**Option 3** - Both MnS consumer (SMO) and producer (NF) support REST/HTTPS as a common basic mechanism of Notification service to guarantee compatibility, and Kafka is treated as an alternative mechanism.

## 7.4 Trace Transport

### 7.4.1 Overview

The transport of trace specified in O1 specifications is file-based (bulk) and/or streaming (real time).

File-based trace collects trace records in files that are available to the consumer with a time delay. The producer optionally sends the file directly to the consumer or sends the location of the file to the consumer.

Procedures for naming the trace data file are described in 3GPP TS 32.423[i.27] clause B. File Naming Convention is described in 3GPP TS 32.423[i.27] clause B.1. Trace files are produced in XML format. The XML format is described in 3GPP TS 32.423[i.27] clause A2.2. Example XML files are provided in 3GPP TS 32.423[i.27] Annex D.

In the case of streaming trace, the data is sent in bursts across a WebSocket connection to the consumer, maintaining the relevance of the data while minimizing transport overhead.

Definitions of trace record content for all trace types, XML trace file format, and streaming trace GPB record definition are found in 3GPP TS 32.423[i.27]. definitions for streaming data reporting are specified in 3GPP TS 28.532[i.3].

For trace data streaming, A secure WebSocket connection is established between the Trace MnS Producer and the Trace MnS Consumer. The description of trace data streaming is in clause 7.1.3 Transport of Traces in Current SBMA Solution Sets of the present document.

### 7.4.2 Issue Summary

The issues of WebSocket for trace data streaming are as in clause 7.2.2 issue summary of WebSocket for performance data streaming of the present document.

### 7.4.3 Potential solution

Potential solution of Trace transport protocol is as in clause 7.2.3.



## Annex A:

### Change history/Change request (history)

| Date       | Revision | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023.04.19 | 00.00    | Skeleton Document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2023.05.10 | 00.01    | Incorporated MAV-2023.04.19-WG10-CR-0012-CNOAM-Outline-v01.docx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2023.06.14 | 00.02    | Incorporated MAV-2023.05.03-WG10-CR-0013-CNOAM-DataFormat-Outline-v04.docx<br>Incorporated MAV-2023.05.03-WG10-CR-0014-CNOAM-DataManagementOperations-Outline-v05.docx<br>Incorporated MAV-2023.05.03-WG10-CR-0015-CNOAM-TelemetryOperations-Outline-v04.docx<br>Incorporated MAV-2023.05.03-WG10-CR-0016-CNOAM-TelemetryTransport-Outline-v04.docx                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2023.07.19 | 00.03    | Incorporated <a href="#">MAV-2023.06.19-WG10-CR-0017-CNOAM-ComponentB-v03.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2023.09.20 | 00.04    | Incorporated <a href="#">MAV.AO-2023.09.06-WG10-CR-0043-CNOAM-Introduction-v03.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2023.11.14 | 00.05    | Incorporated <a href="#">MAV-2023.06.19-WG10-CR-0018-CNOAM-Non3GPPIMDM-v07.docx</a><br>Incorporated <a href="#">MAV-2023.06.19-WG10-CR-0019-CNOAM-ComponentA-v04.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2023.11.29 | 00.06    | Incorporated <a href="#">MAV-2023.08.09-WG10-CR-0034-CNOAM-TelemetryTransport-v04.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2023.12.13 | 00.07    | Incorporated <a href="#">ERI-2023.08.09-WG10-CR-0045-CNOAM-MnSComponentB-v05.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2023.01.31 | 00.08    | Incorporated <a href="#">MAV-2023.12.07-WG10-CR-0027-CNOAM-PMEEventsTraces-v05.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2024.05.02 | 00.09    | Incorporated CMCC-2024.01.24-WG10-CR-001-AddReference-v02.docx<br>Incorporated CMCC-2024.01.24-WG10-CR-002-AddConsumers of Configuration Data-v05.docx<br>Incorporated MAV-2024.02.15-WG10-CR-0020-CNOAM-DeclarativeImperative-v03.docx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2024.06.06 | 00.10    | Incorporated ERI-2024.05.21-WG10-CR-0079-CNOAM-Replace requirements clause names-v01.docx<br>Incorporated MAV-2024.04.11-WG10-CR-0021-CNOAM-DeclarativeImperative-IssueSummary-v05.docx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2024.06.07 | 00.11    | Incorporated MAV-2024.05.16-WG10-CR-0022-CNOAM-DeclarativeImperative-NewRequirements-v03.docx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2024.08.28 | 00.00.12 | Incorporated <a href="#">CMCC.AO-2024.05.22-WG10-CR-005-Consumers of Configuration Data-add issue summary-v06.docx</a><br>Incorporated <a href="#">MAV-2024.06.06-WG10-CR-0023-CNOAM-Transport-of-Events-in-Current-Solution-Sets-v06.docx</a><br>Incorporated <a href="#">MAV-2024.06.06-WG10-CR-0024-CNOAM-Add-Term-Telemetry-v03.docx</a><br>Incorporated <a href="#">MAV-2024.06.11-WG10-CR-0025-CNOAM-Transport-of-Configuration-Management-Operations-v06.docx</a><br>Incorporated <a href="#">CMCC.AO-2024.06.11-WG10-CR-006-Merge532and533-v05.docx</a><br>Incorporated <a href="#">MAV-2024.06.11-WG10-CR-0026-CNOAM-Source-of-Truth-v05.docx</a><br>Incorporated <a href="#">MAV-2024.07.18-WG10-CR-0029-CNOAM-Source-of-Truth-Issue-Summary-v02.docx</a> |
| 2023.10.17 | 00.00.13 | Incorporated <a href="#">MAV-2024.09.19-WG10-CR-0030-CNOAM-Transport-of-Traces-in-Current-Solution-Sets-v01.docx</a><br>Incorporated <a href="#">MAV-2024.10.03-WG10-CR-0031-CNOAM-Configuration-Repository-Reference-Update-v01.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2024.12.19 | 00.00.14 | Incorporated <a href="#">ERI-2024.10.21-WG10-CR-0131-CNOAM-Editorial clean up clause 5.3.2-v01.docx</a><br>Incorporated <a href="#">MAV-2024.10.10-WG10-CR-0033-CNOAM-Performance-Metric-Transport-Overview-v02.docx</a><br>Incorporated <a href="#">CMCC.AO-2024.10.11-WG10-CR-007-addMnSCompnentC-v08.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2025.01.13 | 00.00.15 | Incorporated <a href="#">MAV-2024.12.19-WG10-CR-037-CNOAM-TR-Outline-Update-v01.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2025.03.24 | 00.00.16 | Incorporated <a href="#">CMCC.AO-2025.03.03-WG10-CR-008-addAbbreviation-v02.docx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2025.06.09 | 00.00.17 | Incorporated <a href="#">MAV-2024.07.18-WG10-CR-0028-CNOAM-Transport-of-Config-Mgmt-Operations-Issue-Summary-v10.docx</a><br>Incorporated <a href="#">MAV-2025.04.27-WG10-CR-0038-CNOAM-Trace-Transport-Overview and Issue Summary-v06.docx</a><br>Incorporated <a href="#">MAV-2025.05.19-WG10-CR-0048-CNOAM-Trace-And-Performance-Metric-Transport-Potential-Solution-V04.docx</a><br>Incorporated <a href="#">MAV-2024.10.16-WG10-CR-0035-CNOAM-Notification-Potential-Transport-</a>                                                                                                                                                                                                                                                                            |

|          |          |                                                                                                                                                                                                                                                                                                                       |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |          | <a href="#">Issue-And-Potential-Solution-v12.docx</a>                                                                                                                                                                                                                                                                 |
| 20250625 | 00.00.18 | Incorporated <a href="#">ERI-2025.06.11-WG10-CR-0179-CNOAM-Editorial clean up</a> , <a href="#">ERI-2025.06.11-WG10-CR-0180-CNOAM-Scope</a> , <a href="#">CMCC-2025.06.13-WG10-CR-009-Removal+of+not+used+informative+references</a> , <a href="#">CMCC-2025.06.13-WG10-CR-010-Correction+of+mistaken+reference</a> , |
| 20250710 | 00.00.19 | Incorporated <a href="#">CMCC-2025.06.27-WG10-CR-011- Editorial-clean-up-v04.docx</a>                                                                                                                                                                                                                                 |